

Rohan Chitnis

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Education

Massachusetts Institute of Technology, Cambridge, MA
GPA: 4.8 / 5.0. Area: Robotics and Machine Learning. Minor: Econometrics.
Doctor of Philosophy in Electrical Engineering and Computer Science, May 2022.
Master of Science in Electrical Engineering and Computer Science, May 2018.
University of California, Berkeley, Berkeley, CA
Graduated with Highest Honors (GPA in top 3%). GPA: 3.95 / 4.0.
Bachelor of Science in Electrical Engineering and Computer Sciences, May 2016.

Experience

Meta Reality Labs, Augmented Reality (AI Research Scientist) 08/22 - present

- Enabling next-gen sEMG wearables via Transformers for timeseries signal processing.
- Researching new frontiers using generative diffusion models and reinforcement learning.

MIT Learning & Intelligent Sys. Lab (Adviser: Leslie Kaelbling) 09/16 - 08/22

- Conducted research in artificial intelligence for robotics, toward a PhD.
- Focused on integrated learning and planning for solving long-horizon robotic tasks.

Nuro Inc., Planning (Software Engineering Intern) 06/21 - 09/21

- Explored integrated data-driven and rule-based reasoning to improve planning stack.
- Developed a synergistic approach that improves behavior in 20-40% of challenge scenes.

Facebook AI Research, Robotics (Research Intern with Abhinav Gupta) 06/19 - 09/19

- Researched intrinsic motivation for synergistic behavior via deep reinforcement learning.
- Built and tested my algorithms in real-world bimanual manipulation environments.

Google Brain, Robotics (Research Intern with Sergey Levine) 05/17 - 09/17

- Researched methods for speeding up deep reinforcement learning for robotics, by regressing on event outcomes to improve exploration.

Airbnb Inc., Search Ranking (Software Engineering Intern) 06/16 - 08/16

- Incorporated mobile data into search ranking ML models.

UC Berkeley Robot Learning Lab (Adviser: Pieter Abbeel) 02/13 - 05/16

- Performed work in hierarchical combined task and motion planning.
- Lead coordinator of lab outreach program, providing tours to visitors of varied ages.

Selected Publications

IQL-TD-MPC: Implicit Q-Learning for Hierarchical Model Predictive Control. Rohan Chitnis*, Yingchen Xu*, Bobak Hashemi, Lucas Lehnert, Urun Dogan, Zheqing Zhu, Olivier Delalleau. ICRA 2024.

Learning Neuro-Symbolic Relational Transition Models for Bilevel Planning. Rohan Chitnis*, Tom Silver*, Joshua Tenenbaum, Tomás Lozano-Pérez, Leslie Pack Kaelbling. IROS 2022.

Planning with Learned Object Importance in Large Problem Instances using Graph Neural Networks. Tom Silver*, Rohan Chitnis*, Aidan Curtis, Joshua Tenenbaum, Tomás Lozano-Pérez, Leslie Pack Kaelbling. AAAI 2021.

Intrinsic Motivation for Encouraging Synergistic Behavior. Rohan Chitnis, Shubham Tulsiani, Saurabh Gupta, Abhinav Gupta. ICLR 2020.

Honors / Awards

Facebook Fellowship Finalist (Machine Learning track), 2020. Top 4% of applicants.
NSF GRFP Fellow, 2016. Awarded NDSEG Fellowship (declined).
Hertz Fellowship Finalist, 2016. One of 40 finalists nationwide.
Runner-up for the CRA Outstanding Undergraduate Researcher Award, 2016.
Sole recipient of the EECS Mark D. Weiser Excellence in Computing Scholarship, 2015.
UC Berkeley Outstanding Graduate Student Instructor (OGSI) Award recipient, 2015.
UC Berkeley Regents' and Chancellor's Scholar.

Technical Skills

Languages: Python, C++, Java, JavaScript, LaTeX.
Software: PyTorch, TensorFlow, Theano, Unix/Linux, Robot Operating System (ROS), OpenCV, Apache Pig, Apache Spark, Hadoop MapReduce.